

Open Horizons, Colliding Bodies: Adaptive Lure Agents in a Glass Maze

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Figure 1: Installation View of the Artwork.

Abstract

Why build yet another virtual reality artwork when VR already promises boundless exploration? We began with a blunt provocation: make the body feel the gap between virtual promise and material fact. Our installation embeds a fully occlusive headset inside a transparent glass maze so constraint remains visible. An adaptive reinforcement-learning (RL) lure, treated as a performing

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partner—edits salience with gentle invitations rather than commands. Across over ten exhibitions (2017–2025), the piece evolved from onlooker voyeurism to a choreography of agency among audience, machine-performer, and artist-architect. By partially publicizing policy intent on the floor, projections, and rail, what is usually hidden as seamless UX becomes dramatically legible; guidance is experienced as both bodily and social, with collisions reading as shared mischief more than punishment. We contribute an art-led, large-scale architecture that couples predictive models with embodied interaction while retaining ambiguity: visible constraint (glass), public policy signals (floor/projections/rail), and an RL lure that persuades by staging opportunity. Drawing on in-exhibition observations, we report regularities that motivate lightweight, gallery-feasible evaluations without altering the artwork, thereby paving the way for systematic study. More broadly, the work reframes “good guidance” not only as seamlessness but as strategic visibility, an ethico-aesthetic stance on when power should be allowed to

show, and the autonomy we trade when an algorithm anticipates our limits.

CCS Concepts

• **Applied computing** → **Media arts**; • **Human-centered computing** → *Virtual reality*; • **Theory of computation** → Multi-agent reinforcement learning.

Keywords

Embodied Interaction, Algorithmic Persuasion

ACM Reference Format:

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1 Introduction

VR pursues what Bolter and Grusin call *transparent immediacy*: traversing infinite worlds as the medium recedes [Bolter & Grusin 1999]. We use VR for fully occlusive headset worlds, and XR for the reality - virtuality continuum [Milgram et al. 1994; Skarbez et al. 2021]. Early landmarks, Char Davies’s *Osmose* (1995) and Jeffrey Shaw’s pedal-driven *Legible City* (1989), translated breath or effort into boundless landscapes, yet remained indifferent to travelers [Davies 1995; Shaw & Groeneveld 1989].

Locomotion research algorithmically masks spatial limits. Redirected walking subtly warps visuals to keep users circling in small play areas [Razzaque 2005]. New controllers predict footsteps and pre-rotate scenery, including multi-user variants [Hirt et al. 2024]. Reinforcement learning also steers people to chosen locations [Rovira & Slater 2017]. Viewed through Fogg and Zuboff, these systems capture micro-behaviors to shape the *next* action [Fogg 2003; Zuboff 2019]. Framed as convenience, such guidance still imposes bodily demand; embodiment studies show that threats to a virtual body elicit physiological responses [Slater et al. 2010].

Media arts precedents foreground mediation and spectatorship: Blast Theory’s *Desert Rain* (1999) plays behind a water screen [Blast Theory 1999]; Dan Graham’s mirrored pavilions fold spectatorship back on itself [Enright & Walsh 2009]; Lozano-Hemmer’s *Pulse Room* (2006) maps heartbeats to light [Lozano-Hemmer 2006]. Yet such work rarely *recompute* in real time to steer whole bodies. In live art and HCI, machine learning, whether for interactive art engagement [Tong & Kulić 2021] or for social-harmony agents [GuestXR Consortium 2025], tends to prioritize comfort rather than expose control. We position this work as research through art practice, grounded in longitudinal *observational* evidence from ten exhibitions; our lightweight study plans scaffold future inquiry rather than mandate methods, preserving the artwork’s dramaturgy.

Our installation couples a rigid glass grid with an adaptive RL lure that replots invitations around each visitor. The next section shows how this pairing materializes whole-body algorithmic persuasion and why it matters for agency. We surface and socialize predictive guidance rather than hide it: constraint stays visible, and aspects of policy intent are public. We treat the RL policy as a performing partner whose tactics are composed and scored as dramaturgy, not tuned solely for efficiency.

Table 1: Positioning within XR and media arts. An adaptive RL lure is paired with a rigid glass grid; rather than hiding constraints (RDW) or relying on invisible guidance (ML/XR), persuasion is staged and parts of policy state are public.

Approach	Typical goal	Mechanism (examples)	Relation to this work
<i>Redirected Walking (RDW)</i>	Fit large VEs into small rooms	Gains/curvature; predictive and multi-user controllers [e.g., Razzaque 2005; Hirt et al. 2024]	Constraints stay visible: a rigid glass grid while invitations shape motion.
<i>RL/ML guidance in XR</i>	Nudge toward targets or social harmony	RL steering to locations; social-harmony agents [Rovira & Slater 2017; GuestXR Consortium 2025]	RL lure is <i>theatricalized</i> —persuasion is visible/legible to onlookers.
<i>Media arts legibility</i>	Expose systems and spectatorship	Water screens; mirrors; physiological mapping [Blast Theory 1999; Enright & Walsh 2009; Lozano-Hemmer 2006]	Extends legibility to agent <i>policy state</i> (rail / floor / projections); probes the “legibility paradox.”

Contributions. (1) A material-computational coupling (glass grid + adaptive RL lure) that keeps constraint visible while making policy intent publicly readable; (2) observational evidence from ten exhibitions laying the groundwork for future studies; (3) a framing of *strategic visibility* and the *legibility paradox* that repositions guidance in XR from hidden convenience to public negotiation.

2 Motivation and Concept

Our inquiry began with a blunt provocation: how can the body *feel* the gap between virtual promise and material fact? In the 2017 premiere of *Do Androids Dream of Electric Cows?*, a transparent glass maze made that gap bodily obvious: visitors saw an endless digital plain yet collided with solid panes. With spectators ringed around the maze, each stumble was public, folding desire and scrutiny into one scene. Agency first appears as friction between visual freedom and bodily arrest; this paper traces how that tension deepened across ten exhibitions. We keep constraint visible and algorithmic intent legible, turning predictive guidance into a body scale, a spectated encounter. Collisions are staged as playful, shareable mischief rather than punitive failure, making impact a social beat that binds wearer and witnesses.

Across the exhibitions, the scene is a trio: audience, machine-performer, artist-architect. Audience: eyes chase openness; the body meets glass; and just when the LED floor, projections, and overhead rail seem to explain the system, a lure overturns that “mastery”. Machine: a reinforcement-learning lure profiles gait and holds rewards just out of reach. Artist: an uncompromising material grid makes missteps felt, not abstracted. Authority is braided and unbraided each time someone steps inside.

A complementary tactic sustains desire. At the VR horizon’s *center* glows the never-depicted Tannhäuser Gate; lacking a canonical image, it remains a receding vector of longing. Existing only in VR, each unseen collision exacts a toll for pursuing a promise that never settles.



Figure 2: Overall Installation View: Audiences navigate the maze while wearing VR headsets.

Technology follows intent. Crystal-clear glass forms an invisible yet absolute boundary; the RL lure personalizes guidance. Outside the maze, an LED floor, pane projections, and an addressable rail externalize parts of the policy state. These cues invite spectators to think they “understand” the system, yet inside the headset that legibility falters. Each layer responded to field observation: when laughter dominated, we increased transparency; when tiptoeing emerged, we trained the lure to restore momentum. In short, every addition heightens a single drama, an ongoing negotiation over who, or what, steers the body. These are aesthetic as much as technical choices: glass as the politics of optical promise; floor reflections as scenography; pane “windows” as glimpses of machine attention; the rail as a concise notation of policy state.

The installation turns otherwise invisible guidance into a shared, discussable experience that persists beyond the gallery.

Taken together, these choices motivate our contribution to XR and media arts. We present a design that keeps physical constraint visible while making algorithmic intent publicly legible: an unyielding glass grid paired with an adaptive RL lure whose policy is partially externalized via floor, projections, and rail. This coupling renders persuasion bodily and social: visitors feel guidance at whole-body scale while bystanders read, and sometimes misread,

the system, a legibility paradox. Within this dramaturgy we document lure behaviors *in exhibition* and the resulting choreography of agency among audience, machine, and artist, offering a transferable pattern for XR artworks that expose rather than conceal control. Because the system is instrumented and has run across ten exhibitions, it also serves as a testbed for lightweight evaluations that formalize these experiential claims without altering the artwork.

3 Artwork Appearance

To keep constraint visible and intent public, the installation evolved through three material iterations that balance embodied tension with spectator legibility. The initial edition used a 10×10 m matrix of 42 glass panes in stainless rails. Under raking light, reflective edges lured with transparency yet blocked movement, producing palpable disorientation.

To externalize the private struggle without dissolving it, we added a floor-scale LED display. At ground level it reads as a reflecting pool beneath a floating horizon; from above, reflections merge into a coherent landscape that grants spectators a commanding vantage. Pane collisions trigger concentric ripples, converting a wearer’s frustration into visible pulses while preserving the necessity of the headset for continuous immersion.

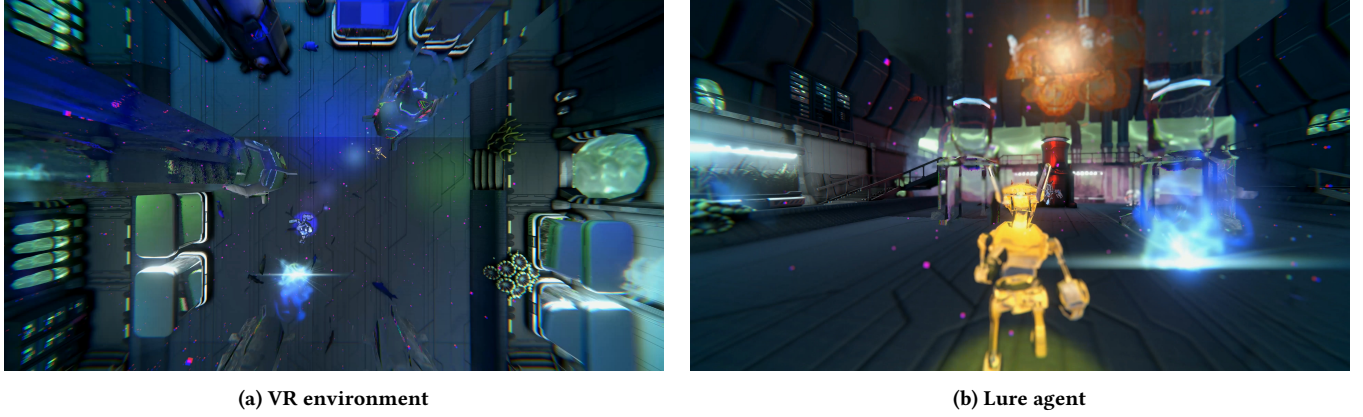


Figure 3: VR environment (left) and the lure agent (right).



Figure 4: From above, the LED floor functions as a reflecting pool, mirroring the virtual world within the physical maze.

We then introduced selective projection onto interior panes, overlaying virtual fragments on the physical grid. These “windows” heighten spatial ambiguity and hint at a roaming viewpoint without making the mechanism explicit.

Finally, an anodized aluminum *LED rail* along the top edge expresses system intent as running pulses with varying intensity. The aim is not to guide spectators but to make the model-participant negotiation legible at a glance. Within VR, crystalline lure agents glow and gesture to attract attention; outside, the floor, projections, and rail translate private bodily tension into visible choreography.

Together these elements use material clarity (glass), spatial reflection (floor), partial revelation (projections), and succinct signaling (rail) to stage persuasion as both embodied and public. They set up the mechanisms in Section 4, where we describe how policy state is computed and routed.

4 System Design

To make constraint visible and policy legible, the system couples lean sensing with an adaptive policy and two coordinated channels: one for the wearer in VR, one for spectators. The *glass grid* is the

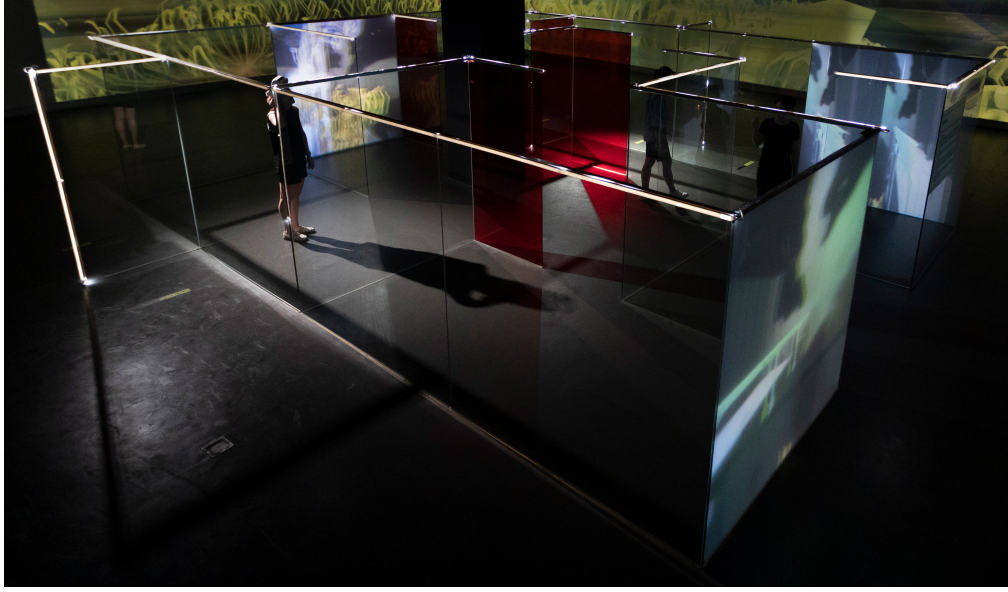


Figure 5: Pane projections open “windows” into the VR scene; the overhead rail renders aspects of policy intent legible.

non-negotiable substrate: it fixes absolute geometry the policy must respect and keeps constraint materially visible. Building on this, the RL lure policy modulates salience rather than commanding action, so voluntary movement adheres to an unseen plan; in parallel, spectator displays render portions of that plan legible.

Table 2: Surfaces, signals, and perceived effects.

Surface / Agent	Signal source	What viewers perceive
LED floor	Top-view virtual camera; pane-contact events	Reflecting-pool overview; ripples at collisions (social legibility of frustration)
Pane projections	Roam-cam feeds from lure agents; scheduled reveals	Intermittent “windows” that hint at policy attention without full logic
LED rail	Policy direction vector; confidence scalar; low-pass smoothing	Running pulses (direction) and intensity (confidence)
Lure agents (VR)	RL lure policy outputs (glide, pause, bow); spatialized whisper	Slow glides and micro-gestures that invite rather than instruct
Glass grid	Physical substrate; fixed geometry	Non-negotiable constraint, materially visible

Sensing to Policy. Ceiling-mounted *SteamVR Lighthouse* base stations track head pose across the maze. A *tether-free* headset streams recent kinematics (speed, swing, dwell) to the Unity based RL policy. Each frame the policy emits: (1) a target for *lure agents* in VR (glide, pause, bow); (2) channel and gain for a faint spatialized whisper from a compact nearfield speaker ring; and (3) a direction vector plus confidence for spectator displays. Latency and low-pass smoothing align motion, audio, and light as one cue.

Wearer channel: glass and lure agents. The headset presents an open horizon anchored by the Tannhäuser Gate; the glass grid enforces boundaries the eyes cannot see. Within that field, crystalline lure agents realize policy invitations through slow glides and differential micro gestures that read as opportunity rather than instruction. The whisper reinforces direction without speech, sustaining voluntary discovery. The result is a negotiation between

visual promise and material fact: the policy edits salience; the glass answers with certainty.

Spectator channel: LED floor, pane projections, and rail. The three surfaces introduced in Section 3 are driven by policy signals as follows. *LED floor*: mirrors a top-view virtual camera; pane-contact events trigger concentric ripple shaders. *Pane projections*: periodically show roam-cam feeds from lure agents as rotating “peek windows.” *LED rail*: maps the direction vector to a running pulse and confidence to intensity, updated with low-pass smoothing. Together, these displays let bystanders read intent at a glance yet remain only loosely helpful once the headset is on.

Closed loop, training, and safety. The loop is head pose informs the policy; the policy updates the VR agent, whisper, and displays; their feedback produces a new pose. The policy is trained in a virtual twin of the maze with rewards for deepened exploration and near-pane approaches while avoiding unsafe states. We freeze weights for exhibitions and run brief daily warm-ups to accommodate drift. Physical constraints stay absolute; turn rates are capped for comfort; staff supervision and soft treatments at critical seams mitigate risk during crowded runs.

Instrumentation and observed behaviors. Without altering the artwork, we log timestamped head pose, policy outputs (direction/confidence), LED-rail states, pane collision/ripple triggers, and active lure-camera selections. These traces enable lightweight toggles (rail/projections on-off; lure enabled-disabled) in future evaluations. In exhibition we consistently observe three persuasive behaviors: a gentle *horizon drift* during steady strides; a *false exit* during stalls; and a *peripheral siren* that seeds motion at the edge of vision. We report these as field observations rather than controlled effects.

5 Observational Findings

We report patterns that clarify how the artwork’s dramaturgy unfolds in exhibition, prior to formalizing them as hypotheses. Across ten exhibitions we drew on two sources: curator/artist field notes and system logs—timestamped head-pose series, pane collision and ripple events, policy direction and confidence, LED-rail states, and active lure-camera selections (Section 4). These are descriptive regularities *in exhibition* that guided iteration; they are not controlled effects but motivate the hypotheses in Section 6.

Visitors often start boldly, pause after the first bump, then re-accelerate as the lure restores momentum (H1). Cautious tiptoeing slows progress but does not reliably reduce contacts (H2). Subtle peripheral invitations (edge-of-vision agent motion plus whisper) bend paths that feel “straight” to the wearer (H3). Spectators form confident theories from the floor, projections, and rail that fail to transfer in headset—the *legibility paradox* (H4). Venue factors (brighter light, larger rooms) shift spectator affect more than wearer outcomes; hesitation persists (H5). Collisions are often received as *shared mischief* rather than punishment, especially when bystanders audibly anticipate impacts (H6).

These observations support three shifts to common XR/HCI assumptions: (1) when *constraint stays visible* and *policy intent is partly public*, guidance becomes *shared* bodily for wearers and socially legible (sometimes over-legible) for spectators; (2) partial legibility induces bystander overconfidence that does not transfer in headset (the *legibility paradox*); and (3) subtle *peripheral* invitations can bias whole-body curvature while preserving a sense of autonomy. Together, these claims counter the usual emphasis on concealed constraint and seamless guidance. H1-H6 in Table 3 are posed as testable hypotheses using the existing installation and logs, without altering the artwork.

6 Limitations And Future Work

Our stance is observational and in-exhibition. Across venues the installation produced traces from existing instrumentation head-pose series, pane collision/ripple events, policy direction/confidence, LED-rail states, and lure-camera selections (Section 4). Conditions vary, and the RL policy was tuned for persuasiveness *in situ*, not laboratory control; read this account within those bounds. Even so, the installation is a ready testbed for systematic evaluation without altering the work.

Building on this, we propose IRB-feasible, concise studies that preserve dramaturgy while substantiating H1-H6. Short within-subject blocks alternate the RL lure *enabled vs. disabled* and displays *visible vs. muted*, plus minimal manipulations: *whisper gain low vs. normal* and *reduced vs. normal peripheral placements*. Using existing logs, compute time to first pane, post-collision recovery, hesitation episodes, path curvature, speed variance, and near-pane approaches, paired with one line exit prompts on guidance and agency. Outside-view vignettes test bystander legibility: observers predict lure moves or wearer paths with displays visible or muted; track accuracy and confidence over repeated exposures to probe the paradox. sub-minute presence/agency items and optional 30 s voice notes probe embodiment and control; dual-wearer sessions

Table 3: Each hypothesis is grounded in current observations and paired with a minimally intrusive next test.

ID	Observed regularity	Field notes and log signals	Future evaluation
H1	Bold start → pause after first bump → recovery with the lure	Head-pose speed series; timestamp of first pane collision; notes on slowdown and re-acceleration	Toggle RL lure <i>enabled vs. disabled</i> ; compare post-collision recovery time/speed; report time to first pane
H2	Slow, cautious gait reduces speed but not collisions	Head-pose speed minima; collision counts per minute; near-pane approach counts; notes	Segment sessions by speed (no instruction change); compare collision rate and near-pane approaches across slow vs. normal segments
H3	Peripheral invitations bend paths that feel “straight”	Path curvature vs. head yaw; lure-camera selections; wearer reports (“I walked straight”)	Toggle whisper gain <i>low vs. normal</i> ; reduce vs. keep peripheral lure placements; compare path curvature and heading error
H4	Bystanders’ confident theories outside do not transfer inside (legibility paradox)	Spectator remarks; wearer reports after exit; mismatches between predicted and experienced behavior	Bystander prediction task with <i>displays visible vs. muted</i> ; measure path/lure-move prediction accuracy and confidence over repeated exposures
H5	Venue light/size changes tone more than outcomes	cross venue normalized collision/hesitation metrics; notes on ambient light and room size	Compare core metrics across venues; correlate with ambient light level and room size (observational)
H6	Collisions are framed as shared mischief more than punishment	Field notes and crowd audio cues (anticipatory laughter); brief post-wearer comments	one line exit sentiment prompt (mischief ↔ punitive); analyze relative to collision density

and crowd heatmaps extend to second-order persuasion, with multi-agent arbitration a natural next step. Traces are de-identified; logging is disclosed with opt-outs; accessibility is supported via tactile edge cues or staff guided alternatives. The program deepens rather than instrumentalizes the dramaturgy—measuring without undoing the conditions that make the negotiation felt and public.

7 Conclusion

This work stages guidance as a public negotiation rather than a hidden convenience. An unyielding glass grid keeps constraint perceptible; an adaptive RL lure edits salience through invitations; and policy state is surfaced to spectators via the floor, pane projections, and the overhead rail. In exhibition, collisions become anticipatory, comic beats shared by onlookers, and public signals foster confidence that does not fully transfer in headset—the legibility paradox.

Beyond a single piece, the installation functions as an art-led research instrument. It offers a replicable pattern for coupling predictive models with embodied interaction while preserving ambiguity, and it provides a ready, minimally intrusive testbed for systematic study (H1-H6). We advocate *strategic visibility*: making

certain forms of power legible at the moment they shape action, and measuring how autonomy is perceived when guidance is felt rather than commanded.

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